

Software Requirements Specification

Sakai LMS — Video Training Module

Version 3.0

March 2026

Revision History

v1.0 — 18 Mar 2026 — Initial draft from 7 stakeholder requirements

v2.0 — 18 Mar 2026 — Stakeholder review #1; added Private Channel Storage, External LMS API

v3.0 — 18 Mar 2026 — All open questions resolved; API refined to read + manage (no create);
platforms finalized to YouTube + Vimeo; student uploads excluded; max 20 API clients

CONFIDENTIAL

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines the functional and non-functional requirements for a new Video Training Module to be integrated into the Sakai Learning Management System (LMS). The module introduces a centralized, role-aware video management and consumption experience for instructors and students, addressing a gap that existing Sakai tools (Resources, Lessons, Kaltura integration) only partially cover.

Version 3.0 incorporates all remaining stakeholder decisions: the External LMS API is finalized as read + manage (no video creation via API); supported external platforms are YouTube and Vimeo only; student-generated video uploads are excluded from scope; maximum 20 simultaneous external API clients; REST is the sole API protocol; and no multi-instance federation is required.

1.2 Scope

The VTM will be delivered as a new Sakai tool accessible at institution, course, and lesson levels. It encompasses video upload (local and to private channels on YouTube/Vimeo), external link embedding, metadata management, categorization, visibility control, cross-referencing between the video library and Lessons, playback analytics, accessibility compliance, and a RESTful API enabling up to 20 connected external LMS platforms to query the video catalog and manage existing videos.

Explicitly out of scope: student-generated video uploads, live streaming, DRM, GraphQL API, and multi-instance federation.

1.3 Definitions and Acronyms

Term	Definition
VTM	Video Training Module — the system being specified
Global Video Page	Institution-wide public video catalog visible to all authenticated Sakai users
Course Video Page	Per-course video library visible to enrolled students and course staff
Lesson Video	A video embedded within a specific Lesson page via the Lessons tool
Visibility Scope	The audience setting of a video: Global, Course, or Lesson-only
Private Channel	An unlisted or private channel/playlist on YouTube or Vimeo used as remote storage for VTM videos
External LMS	A third-party Learning Management System that integrates with Sakai via the VTM API to access and manage videos
Instructor	A user with the Instructor or Teaching Assistant role in a Sakai site
Student	A user with the Student / Participant role in a Sakai site
Admin	A user with Sakai system-level administrative privileges
LTI	Learning Tools Interoperability — standard for integrating external tools
oEmbed	Open format for embedding content from a URL, used by YouTube, Vimeo, and others
xAPI	Experience API (Tin Can) — specification for learning analytics data collection

1.4 Stakeholders

- Instructors / Course Designers — create, manage, and publish video content
- Students / Participants — consume video content, track progress (no upload capability)
- System Administrators — configure quotas, manage global library, enforce policies, manage API clients
- External LMS Administrators — consume the VTM API to query and manage video catalogs (up to 20 clients)
- Accessibility Officers — ensure WCAG 2.1 AA compliance
- IT / DevOps — institutional media server, CDN, transcoding pipelines, API gateway

1.5 All Stakeholder Decisions (Consolidated)

The following table consolidates all stakeholder decisions from v1.0 through v3.0. These are binding design decisions referenced throughout this specification.

ID	Question	Answer	Impact on Specification
SD-1	Preferred storage backend	Institutional media server + YouTube/Vimeo private channels	Architecture uses institutional server as primary; YouTube/Vimeo as optional secondary
SD-2	Auto-captioning service	External third-party API (Google, AWS Transcribe, Whisper)	VTM-FR-075: captioning bridge with API key management
SD-3	DRM required?	No. Signed-URL protection sufficient	NFR-041 simplified: no DRM; signed time-limited URLs only
SD-4	Live streaming?	On-demand only	Scope excludes live/real-time streaming
SD-5	Expected catalog size	Large (100,000+ videos, high growth)	Search indexing mandatory; pagination enforced; performance NFRs tightened
SD-6	Video comments?	Not at the beginning; future phase	Deferred to Phase 2; placeholder VTM-FR-095 at P4
SD-7	Bulk upload?	Deferred to later phase	VTM-FR-011 moved to P4 (Future Phase)
SD-8	Migration tooling?	Optional	Section 8 marked optional; VTM-FR-200 at P3
SD-9	Sakai version	25.x or later	Leverages Sakai 25 APIs and Morpheus 2 portal
SD-10	Student-generated uploads?	No	Students cannot upload videos; upload restricted to Instructor/TA/Admin
SD-11	External platforms for private channels?	YouTube and Vimeo only	Adapters built for YouTube Data API v3 and Vimeo API only; Dailymotion/Wistia excluded
SD-12	External LMS API write scope?	Read + manage existing videos; no create	API supports GET, PATCH, PUT for existing videos; no POST /videos; creation is UI-only
SD-13	Max simultaneous API clients?	20	API gateway sized for 20 concurrent OAuth clients; rate limits adjusted accordingly
SD-14	GraphQL support?	No. REST is sufficient	REST/JSON only; no GraphQL endpoints
SD-15	Multi-instance federation?	No	API serves a single Sakai instance only; no cross-instance queries

1.6 References

- Sakai Community Documentation — <https://www.sakailms.org>

- Sakai 25.x Entity Broker / REST API Documentation
- IMS Global LTI 1.3 / Advantage Specification
- WCAG 2.1 Level AA Guidelines
- xAPI (Experience API) / Tin Can specification for learning analytics
- YouTube Data API v3 — <https://developers.google.com/youtube/v3>
- Vimeo API — <https://developer.vimeo.com>
- OAuth 2.0 / OpenID Connect specifications
- OpenAPI Specification 3.1 (Swagger)

2. System Overview and Architecture Context

2.1 High-Level Architecture

The VTM will be built as a native Sakai tool registered through Sakai 25's tool registration mechanism. It exposes three navigational entry points: (a) a Global Video Page accessible from the institution's top-level navigation, (b) a Course Video Page in the left-hand course menu alongside Lessons, Resources, and Gradebook, and (c) inline video embedding within Lessons pages through an enhanced Lessons content picker.

Internally the module comprises: a Video Repository Service (handling metadata and interfacing with the institutional media server), an External Platform Connector Service (managing OAuth credentials and API interactions with YouTube and Vimeo for private channel storage), a Visibility and Access Control Service (resolving scope rules against Sakai's role/permission framework), a Categorization Engine (managing the taxonomy hierarchy), a Third-Party Captioning Bridge (dispatching audio to external speech-to-text APIs), an External LMS API Gateway (exposing RESTful read and management endpoints for up to 20 connected LMS platforms), and a front-end Playback and Management UI built on Sakai's Morpheus 2 portal framework or as a decoupled React SPA.

2.2 Storage Architecture

The institutional media server is the primary storage backend (SD-1). YouTube and Vimeo private channels serve as optional secondary storage targets (SD-11). The VTM database stores only metadata, pointers (media server asset IDs, external platform video IDs), and thumbnail references.

- Upload workflow: VTM front-end uploads to institutional media server (or YouTube/Vimeo via platform API), receives an asset ID, and stores it in VTM metadata.
- Playback workflow: VTM generates a signed, time-limited URL (institutional server) or platform-embedded player (YouTube/Vimeo) and renders it in the VTM player frame.

2.3 Integration Points

- Institutional Media Server — primary blob storage, transcoding, and streaming delivery
- YouTube Private Channels — optional secondary storage via YouTube Data API v3
- Vimeo Private Channels — optional secondary storage via Vimeo API
- Sakai Lessons — bidirectional: VTM videos embedded in Lessons; Lesson-uploaded videos promoted to VTM

- Sakai Gradebook — completion-based grading passback
- Sakai Events / Statistics — analytics data
- LTI 1.3 — embedding third-party video platforms (Kaltura, Panopto, YuJa, etc.)
- Third-Party Speech-to-Text API — auto-caption generation (SD-2)
- xAPI / Learning Record Store — granular playback event tracking
- External LMS API — REST API for up to 20 connected LMS platforms to read catalog and manage existing videos (SD-12, SD-13)

3. User Roles and Permissions

The VTM adopts Sakai's existing role model and adds granular permissions. Per SD-10, students have no upload capability. The API Client column reflects External LMS API access per SD-12.

Permission	Admin	Instructor	TA	Student	API Client
Upload videos (Global)	✓	✓*	✗	✗	✗
Upload videos (Course)	✓	✓	✓	✗	✗
Upload to private channel	✓	✓	✗	✗	✗
Embed external video link	✓	✓	✓	✗	✗
Edit video metadata	✓	✓	✓†	✗	✓‡
Set visibility scope	✓	✓	✓†	✗	✓‡
Promote Lesson video to VTM	✓	✓	✓†	✗	✗
Withdraw / unpublish video	✓	✓	✓†	✗	✓‡
Delete video permanently	✓	✓	✗	✗	✗
Manage categories (Global)	✓	✗	✗	✗	✗
Manage categories (Course)	✓	✓	✗	✗	✗
Manage private channel config	✓	✗	✗	✗	✗
Manage API clients / keys	✓	✗	✗	✗	✗
View analytics (Global)	✓	✗	✗	✗	✗
View analytics (Course)	✓	✓	✓	✗	✓‡
Browse / watch videos	✓	✓	✓	✓	✓
Bookmark / favourite videos	✓	✓	✓	✓	✗
Set storage quotas	✓	✗	✗	✗	✗
Create videos via API	✗	✗	✗	✗	✗

* Subject to admin approval workflow if configured. † Only for videos they uploaded. ‡ Within the API client's authorized scope; manage-only (no create per SD-12).

Δ v3.0: Row 'Create videos via API' explicitly blocked for all roles per SD-12. Student upload row removed per SD-10.

4. Functional Requirements

4.1 Video Upload and Ingestion

Δ v3.0: *Student-generated uploads explicitly excluded (SD-10). Only Instructors, TAs (course-level), and Admins may upload.*

[VTM-FR-010] Direct File Upload

Instructors shall be able to upload video files from their local device. The system shall support at least MP4 (H.264/H.265), WebM, MOV, AVI, and MKV. Files shall be transcoded server-side to H.264 MP4 with AAC audio at configurable quality tiers (360p, 720p, 1080p). The primary storage target is the institutional media server; instructors may alternatively select a configured YouTube or Vimeo private channel (see 4.11).

Priority: Must-have (P1)

Source: Original Req #1

- Maximum file size configurable by admin (default 2 GB)
- Upload shall support chunked/resumable transfers for large files
- Progress indicator displayed during upload and transcoding
- Students cannot upload videos (SD-10)

[VTM-FR-011] Bulk Upload [DEFERRED — Future Phase]

Instructors shall be able to upload multiple videos in a single operation via drag-and-drop or file picker multi-select. Deferred to a future phase per SD-7.

Priority: Future Phase (P4)

Source: Derived — deferred per SD-7

[VTM-FR-012] External Video Link Embedding

Instructors shall be able to add a video to the VTM by providing an external URL. The system shall support oEmbed-compatible sources (YouTube, Vimeo) as well as generic iframe embeds. External videos shall be displayed inline via the embedded player of the source platform.

Priority: Must-have (P1)

Source: Original Req #7

- Auto-detect provider and extract thumbnail, title, duration when available
- Instructors may override auto-detected metadata
- External links validated on save and periodically checked (cron job)
- Broken links trigger instructor notification and are flagged in management UI
- Supported oEmbed providers: YouTube, Vimeo (SD-11); generic iframe for others

[VTM-FR-013] LTI-Based Video Source Integration

The system shall support embedding videos from institutional video platforms (Kaltura, Panopto, YuJa, Ensemble, Mediasite) via LTI 1.3 tool launches. Admin shall configure LTI connections at system level; instructors browse and select content from the connected platform within the VTM content picker.

Priority: Should-have (P2)

Source: Derived — institutional video platform integration

4.2 Visibility and Access Control

[VTM-FR-020] Visibility Scope Assignment on Upload

When uploading or linking a video, the instructor shall select a Visibility Scope: (a) Global — visible on the Global Video Page to all authenticated users, (b) Course — visible only on the Course Video Page to enrolled students and staff, (c) Lesson-only — visible only when embedded inline in a specific Lesson page. Scope applies uniformly regardless of storage location.

Priority: Must-have (P1)

Source: Original Req #1, #3

[VTM-FR-021] Post-Upload Visibility Modification

Instructors shall be able to change Visibility Scope at any time after upload. Scope broadening makes the video appear on the corresponding page immediately (or upon scheduled publish date). Scope narrowing removes the video from video pages but does not break existing Lesson embeds. For Private-Channel-Hosted videos, the system shall update the platform privacy status (see VTM-FR-113). External LMS API clients with manage scope may also change visibility via PATCH endpoint (SD-12).

Priority: Must-have (P1)

Source: Original Req #3

- Scope change requires confirmation dialog when narrowing visibility
- A log entry recorded for each scope change

[VTM-FR-022] Restriction Flag for Lesson-Originated Videos

When a video is uploaded directly within a Lesson page, it defaults to Lesson-only scope. The instructor may promote it to Course or Global scope. A 'Restricted' flag prevents the video from being referenced elsewhere. The External LMS API excludes restricted videos from catalog responses.

Priority: Must-have (P1)

Source: Original Req #2, #3

[VTM-FR-023] Video Withdrawal (Soft Unpublish)

Instructors shall be able to withdraw a video from the video page. A withdrawn video no longer appears in browsing, search results, or API catalog responses, but remains in the system. Withdrawn videos may continue to play in existing Lesson embeds. Withdrawable also via External LMS API PATCH endpoint (SD-12). For Private-Channel-Hosted videos, withdrawal optionally sets external platform video to 'private'.

Priority: Must-have (P1)

Source: Original Req #6

[VTM-FR-024] Scheduled Publishing and Expiry

Instructors may set a publish date and optional expiry date. Before publish date: Draft state, invisible to students. After expiry: auto-transition to Withdrawn.

Priority: Should-have (P2)

Source: Derived — course scheduling alignment

[VTM-FR-025] Prerequisite / Conditional Release

A video's availability on the Course Video Page may be gated by completion of another video or by a Lessons conditional release rule.

Priority: Could-have (P3)

Source: Derived — pedagogical sequencing

4.3 Lesson Integration (Bidirectional)

[VTM-FR-030] Embed VTM Video in Lesson

Instructors shall embed any non-restricted VTM video into a Lesson page. The Lessons content picker includes a 'Video Library' source for browsing/searching both Course and Global libraries. Works seamlessly regardless of storage backend.

Priority: Must-have (P1)

Source: Original Req #4

[VTM-FR-031] Promote Lesson Video to VTM

When a video is uploaded directly in a Lesson, the instructor sees 'Add to Video Library' to copy/link it to Course or Global VTM catalog. Available at upload time and later via context menu.

Priority: Must-have (P1)

Source: Original Req #2, #3

[VTM-FR-032] Lesson Playback Completion Tracking

The system tracks playback progress and reports completion when the student views at least a configurable percentage (default: 90%). Satisfies Lessons completion requirements and optionally passes a grade to Gradebook.

Priority: Should-have (P2)

Source: Derived — learning outcomes tracking

[VTM-FR-033] Reference Indicator

VTM management interface displays a 'Referenced in N lessons' badge. Clicking lists all referencing Lessons with direct links. Also exposed in API video detail response.

Priority: Should-have (P2)

Source: Derived — content management clarity

4.4 Video Metadata and Cataloging

[VTM-FR-040] Core Metadata Fields

Each video shall carry: Title (required), Description (rich text), Tags/Keywords, Duration (auto-detected), Thumbnail URL (auto-generated, custom override), Upload Date, Author/Uploader, Language, Visibility Scope, Status, Storage Location (Institutional/YouTube/Vimeo), External Platform Video ID (if applicable), Course Allocation(s), Category Allocation(s), Restriction Flag. All fields exposed via the REST API.

Priority: Must-have (P1)

Source: Derived — baseline cataloging; expanded for API

[VTM-FR-041] Custom Metadata Fields

Administrators may define additional institution-specific metadata fields (e.g., Department, Academic Year, Content Type). Custom fields included in API responses under a `custom_metadata` object.

Priority: Could-have (P3)

Source: Derived — institutional taxonomy

4.5 Categorization and Organization

[VTM-FR-050] Hierarchical Category Taxonomy

Videos organizable into Category > Subcategory > Group. Admins manage Global taxonomy; Instructors manage course-specific categories. A video may belong to multiple categories.

Priority: Must-have (P1)

Source: Original Req #5

[VTM-FR-051] Course-Based Grouping

On the Global Video Page, videos filterable by originating course. System auto-associates each video with its source course. API supports filtering by `course_id`.

Priority: Must-have (P1)

Source: Original Req #5

[VTM-FR-052] Playlists and Ordered Sequences

Instructors create named playlists within a course. Displayable on Course Video Page and embeddable in Lessons. Students navigate sequentially. Playlists retrievable via API.

Priority: Should-have (P2)

Source: Derived — guided learning paths

[VTM-FR-053] Favourites and Watch Later

Students may mark videos as Favourite or add to Watch Later queue, accessible from 'My Videos' dashboard.

Priority: Should-have (P2)

Source: Derived — student engagement

4.6 Search and Discovery

[VTM-FR-060] Full-Text Search

Keyword search across titles, descriptions, tags, category names, and captions/transcripts. Results respect visibility scope. A dedicated search index (Elasticsearch/OpenSearch) required for the large catalog (SD-5). API exposes identical search via the `q` parameter.

Priority: Must-have (P1)

Source: Derived — discoverability; tightened per SD-5

[VTM-FR-061] Faceted Filtering

UI and API offer faceted filters: Category, Course, Instructor/Author, Duration range, Date range, Language, Tag, Storage Location, and Status (instructor/admin/API only). Combinable with real-time count updates.

Priority: Should-have (P2)

Source: Derived — usability

[VTM-FR-062] Sort Options

Sortable by: Most Recent, Most Viewed, Highest Rated (if enabled), Alphabetical, Duration.
Available in both UI and API.

Priority: Should-have (P2)

Source: Derived — usability

4.7 Video Player and Playback

[VTM-FR-070] Embedded Player Features

For locally hosted videos: adaptive bitrate streaming (HLS or DASH), playback speed (0.5x–2x in 0.25 increments), full-screen, picture-in-picture, volume control, seek bar with preview thumbnails, keyboard shortcuts. For YouTube/Vimeo-hosted videos: platform's native player (YouTube IFrame API, Vimeo Player SDK) embedded within VTM player frame with equivalent controls where supported.

Priority: Must-have (P1)

Source: Derived — modern player expectations

[VTM-FR-071] Closed Captions and Subtitles

Instructors upload caption files (SRT, VTT). Optional auto-generation via third-party speech-to-text API (SD-2). Multiple language tracks supported. For Private-Channel-Hosted videos, captions synced to YouTube/Vimeo when platform API supports caption upload.

Priority: Must-have (P1)

Source: Derived — accessibility (WCAG 2.1)

[VTM-FR-072] Transcript Panel

Synchronized scrolling transcript panel alongside video. Click-to-seek on transcript lines.

Priority: Should-have (P2)

Source: Derived — accessibility and study aid

[VTM-FR-073] Chapter Markers and Timestamps

Instructors define named chapter markers at specific timestamps. Player displays chapter list with direct navigation.

Priority: Should-have (P2)

Source: Derived — long video navigation

[VTM-FR-074] Instructor Annotations / Overlay Notes

Timestamped text notes visible to students during playback as pop-up overlays or side panel. Distinct from captions.

Priority: Could-have (P3)

Source: Derived — active learning

[VTM-FR-075] Third-Party Auto-Captioning Integration

Integrate with external speech-to-text API (Google Cloud Speech-to-Text, AWS Transcribe, or OpenAI Whisper API) per SD-2. Admin configures provider, credentials, default language. System extracts audio, submits to API, ingests VTT result, attaches as auto-generated caption track. Instructor notified on completion.

Priority: Must-have (P1)

Source: New in v2.0 — derived per SD-2

- Admin manages API keys securely (encrypted, not exposed in UI)
- Cost estimation displayed before submitting captioning job
- Queue batches requests and respects API rate limits
- Language auto-detection attempted; instructor may override

4.8 Analytics and Reporting

[VTM-FR-080] View Tracking

Record each playback session: student identity, video ID, start/end time, percentage viewed, playback events. Compatible with Sakai's event system; optionally exported as xAPI. For Private-Channel-Hosted videos, supplement with YouTube/Vimeo Analytics API data where available. Analytics accessible via External LMS API for authorized clients.

Priority: Must-have (P1)

Source: Derived — learning analytics

[VTM-FR-081] Instructor Analytics Dashboard

Per-video metrics: total views, unique viewers, average watch percentage, drop-off heatmap, completion rate. Course-level aggregates also available.

Priority: Should-have (P2)

Source: Derived — instructional improvement

[VTM-FR-082] Student Progress View

Students see their viewing history, completion status per video, and resume from where they left off.

Priority: Should-have (P2)

Source: Derived — self-regulated learning

[VTM-FR-083] Admin Analytics

Institution-level: total storage used (broken down by institutional server / YouTube / Vimeo), most-viewed videos, active courses, bandwidth trends, API usage metrics.

Priority: Could-have (P3)

Source: Derived — capacity planning

4.9 Notifications

[VTM-FR-090] Student Notifications

Via Sakai's notification system: new video published in enrolled course, video approaching expiry, instructor adds annotation.

Priority: Should-have (P2)

Source: Derived — engagement and retention

[VTM-FR-091] Instructor Notifications

Upload/transcode completes or fails, auto-captioning completes or fails, scheduled video auto-publishes, external link broken, Private-Channel-Hosted video becomes unavailable on platform.

Priority: Should-have (P2)

Source: Derived — operational awareness

4.10 Administration and Governance

[VTM-FR-100] Storage Quota Management

Admins set quotas at institution, department, and course levels for the institutional media server. Private-Channel-Hosted videos tracked separately (do not count against local quotas). Warn at 80%, 90%; block uploads at 100%.

Priority: Must-have (P1)

Source: Derived — infrastructure governance

[VTM-FR-101] Bulk Operations

Bulk actions on selected videos: change visibility, assign/remove categories, withdraw, delete, export metadata as CSV.

Priority: Should-have (P2)

Source: Derived — operational efficiency

[VTM-FR-102] Audit Log

All significant actions logged with timestamp, actor, before/after values. Includes: upload, delete, scope change, withdrawal, metadata edit, API key generation, private channel sync, API

management operations. Viewable/exportable by admins. Queryable via External LMS API (admin-scoped clients).

Priority: Must-have (P1)

Source: Derived — compliance and traceability

[VTM-FR-103] Content Moderation Workflow

Optional approval workflow for Global-scope videos. Videos enter 'Pending Approval' state requiring admin approval before appearing on Global Video Page.

Priority: Could-have (P3)

Source: Derived — content quality control

[VTM-FR-104] Retention and Archival Policy

Automatic archival rules: videos not accessed for N months flagged for archival. Institutional server: move to cold storage. Private-Channel-Hosted: update listing status and remove VTM catalog entry. Restorable on demand.

Priority: Could-have (P3)

Source: Derived — storage lifecycle

4.11 Private Channel Storage (YouTube and Vimeo)

Δ v3.0: Platform scope finalized to YouTube and Vimeo only per SD-11. All references to Dailymotion/Wistia removed.

[VTM-FR-110] External Platform Account Configuration

Administrators configure YouTube and/or Vimeo platform accounts at the system level. Configuration includes: platform type (YouTube/Vimeo), OAuth 2.0 client credentials, authorized redirect URI, default channel/user ID, and default privacy setting. Multiple accounts per platform supported (e.g., separate YouTube channels per department).

Priority: Must-have (P1)

Source: Stakeholder Req — Private Channel Storage; refined per SD-11

- OAuth tokens stored encrypted (AES-256) with automatic refresh
- Admin UI: 'Test Connection' button verifies API access
- Each account has a display name for instructor reference

[VTM-FR-111] Upload to Private Channel

When uploading, instructors choose storage target: Institutional Media Server (default) or a configured YouTube/Vimeo private channel. Upload via YouTube Data API v3 resumable upload or Vimeo upload API. Privacy set to 'unlisted' or 'private' as configured. VTM stores external video ID and embed URL.

Priority: Must-have (P1)

Source: Stakeholder Req — Private Channel Storage

- Progress indicator includes external platform upload phase
- Retry up to 3 times on failure; fall back to institutional server
- Storage location indicated by subtle platform icon badge in UI

[VTM-FR-112] Private Channel Playlist Management

Admins map VTM categories or courses to external platform playlists. Auto-add videos to corresponding playlist on upload. Instructors may manually select target playlist.

Priority: Should-have (P2)

Source: Derived — organized external storage

[VTM-FR-113] Privacy Status Synchronization

VTM synchronizes privacy status on YouTube/Vimeo with VTM Visibility Scope. Published → unlisted; Withdrawn/Draft → private. Never sets to 'public' unless explicitly overridden by admin. Async with retry; failures logged and notified.

Priority: Must-have (P1)

Source: Derived — access control consistency

[VTM-FR-114] External Platform Video Import

Instructors import existing videos from configured YouTube/Vimeo channels into VTM without re-uploading. Browsable list via platform API; auto-populate metadata.

Priority: Should-have (P2)

Source: Derived — migration from existing channels

[VTM-FR-115] External Platform Health Monitoring

Periodic verification that Private-Channel-Hosted videos remain available on YouTube/Vimeo. Discrepancies logged, catalog updated, instructor/admin notified. Dashboard shows sync health.

Priority: Should-have (P2)

Source: Derived — operational reliability

[VTM-FR-116] Playback Proxy for Private Channel Videos

Embed YouTube/Vimeo players within VTM player frame for consistent student experience. Direct platform URLs not exposed to students. YouTube: youtube-nocookie.com domain for GDPR. Vimeo: domain-level privacy restriction. Embed tokens generated server-side, short-lived.

Priority: Must-have (P1)

Source: Derived — student experience consistency

4.12 External LMS Integration API

Δ v3.0: *API scope finalized: READ + MANAGE existing videos; NO video creation via API (SD-12). Max 20 simultaneous clients (SD-13). REST only (SD-14). Single-instance only (SD-15).*

[VTM-FR-120] RESTful API Specification

The VTM exposes a RESTful API (OpenAPI 3.1) enabling external LMS platforms to query the video catalog and manage existing video records. The API supports read operations (list, search, get details, get embed URLs) and management operations (update metadata, change visibility, withdraw/republish) but explicitly does NOT support video creation or deletion (SD-12). Versioned at /api/v1/. JSON response format. Single Sakai instance scope (SD-15).

Priority: Must-have (P1)

Source: Stakeholder Req — External LMS API; refined per SD-12, SD-14, SD-15

- OpenAPI spec at /api/v1/openapi.json
- Interactive Swagger UI at admin-configurable path
- No GraphQL endpoints (SD-14)

[VTM-FR-121] API Authentication and Authorization

External LMS clients authenticate via OAuth 2.0 Client Credentials grant. Admins register up to 20 API clients (SD-13), each receiving client_id and client_secret with a permission scope: (a) read-only — catalog queries and embed URL generation, (b) read + manage — additionally allows metadata updates, visibility changes, and withdrawal/republish of existing videos, (c) read + manage + analytics — additionally allows course-level analytics access. No scope permits video creation or deletion. Tokens short-lived (default 1 hour) with refresh.

Priority: Must-have (P1)

Source: Derived — secure integration; refined per SD-12, SD-13

- Each client has a human-readable name, description, contact email
- Clients revocable at any time; revocation invalidates all tokens immediately
- Rate limiting enforced per client (default: 500 requests/minute for read, 60/minute for write)

[VTM-FR-122] Video Catalog Endpoint (Read)

GET /api/v1/videos returns a paginated list of videos accessible to the authenticated client. Each video includes: video_id, title, description, duration_seconds, thumbnail_url, visibility_scope, status, storage_location, upload_date, author, language, tags, categories (with hierarchy), course_allocations (list of course IDs and names), reference_count (number of Lessons embedding it), and caption_tracks.

Priority: Must-have (P1)

Source: Stakeholder Req — External LMS API

- Cursor-based pagination (default page size 50, max 200)
- Filter by: `course_id`, `category_id`, `visibility_scope`, `status`, `tag`, `language`, `uploaded_after`, `uploaded_before`, `storage_location`, `has_captions`
- Sort by: `upload_date`, `title`, `duration`, `view_count`
- Full-text search via `q` parameter

[VTM-FR-123] Single Video Detail Endpoint (Read)

GET `/api/v1/videos/{video_id}` returns full metadata including all FR-122 fields plus: `embed_url` (signed, time-limited), `chapter_markers`, `playlist_memberships`, `lesson_references` (list of Lesson IDs/names), `analytics_summary` (`total_views`, `unique_viewers`, `avg_completion_pct`), and `custom_metadata`.

Priority: Must-have (P1)

Source: Stakeholder Req — External LMS API

[VTM-FR-124] Course Video Listing Endpoint (Read)

GET `/api/v1/courses/{course_id}/videos` returns all videos allocated to a specific course with same fields and filtering/pagination as FR-122.

Priority: Must-have (P1)

Source: Derived — course-centric integration

[VTM-FR-125] Embed URL Generation

POST `/api/v1/videos/{video_id}/embed` generates a short-lived signed embed URL (or HTML snippet). Encodes client ID, expiry, HMAC signature. For Private-Channel-Hosted videos, uses platform's privacy-controlled embed. Default TTL: 4 hours (configurable per client).

Priority: Must-have (P1)

Source: Derived — secure cross-platform playback

- Response includes recommended player dimensions and player type
- Optional `user_id` parameter for per-student view tracking from external LMS

[VTM-FR-126] Video Management Endpoints (Manage)

External LMS clients with 'manage' scope may perform the following operations on existing videos. No video creation or deletion is permitted (SD-12).

Priority: Must-have (P1)

Source: Stakeholder Req — External LMS API; SD-12

The following management endpoints are available:

Method	Endpoint	Description	Required Scope
PUT	/api/v1/videos/{video_id}	Update video metadata (title, description, tags, language, categories, course_allocations, custom_metadata). Cannot change video_id, storage_location, or create new videos.	manage
PATCH	/api/v1/videos/{video_id}/visibility	Change visibility scope (Global, Course, Lesson). Triggers privacy sync for Private-Channel-Hosted videos.	manage
PATCH	/api/v1/videos/{video_id}/status	Change status: publish (Draft→Published), withdraw (Published→Withdrawn), republish (Withdrawn→Published). Cannot set to Archived or Deleted.	manage
PUT	/api/v1/videos/{video_id}/categories	Replace category assignments for a video	manage
PUT	/api/v1/videos/{video_id}/courses	Replace course allocations for a video	manage
PATCH	/api/v1/videos/{video_id}/schedule	Update publish_date and/or expiry_date	manage

Explicitly NOT available via API (SD-12):

- POST /api/v1/videos — video creation is performed only via the Sakai UI by authorized instructors/admins
- DELETE /api/v1/videos/{video_id} — permanent deletion is UI-only (instructor/admin)
- POST /api/v1/videos/{video_id}/captions — caption upload is UI-only
- Any operation that creates new resources (categories, playlists) — UI-only

[VTM-FR-127] Category and Taxonomy Endpoint (Read)

GET /api/v1/categories returns the complete category hierarchy (Global and, if authorized, Course-level). Each category includes: category_id, name, parent_category_id, scope, video_count.

Priority: Should-have (P2)

Source: Derived — taxonomy synchronization

[VTM-FR-128] Playlist Endpoints (Read)

GET /api/v1/playlists and GET /api/v1/playlists/{playlist_id} return playlist metadata and ordered video lists. External LMS may display curated video sequences.

Priority: Should-have (P2)

Source: Derived — playlist integration

[VTM-FR-129] Webhook Notifications

Configurable webhooks push event notifications to registered external LMS endpoints when: a video is published, withdrawn, or has metadata updated; a new video is added to a course; category structure changes. Payloads signed (HMAC-SHA256). Max 3 webhook URLs per client.

Priority: Should-have (P2)

Source: Derived — real-time synchronization

- Admins configure webhook URLs and event subscriptions per client
- Failed deliveries retried with exponential backoff (up to 5 retries / 24 hours)
- Webhook delivery log available to admins

[VTM-FR-130] API Analytics and Usage Reporting

Track API usage per client: total requests, requests by endpoint, error rates, bandwidth consumed. Available to admins via dashboard and exportable as CSV.

Priority: Should-have (P2)

Source: Derived — API governance

[VTM-FR-131] API Health and Status Endpoint

GET /api/v1/health returns API status, Sakai instance version, VTM version, and media server connectivity status. Public (no auth required). Used for monitoring by external LMS clients.

Priority: Must-have (P1)

Source: Derived — operational monitoring

4.13 Deferred Features (Future Phase)

[VTM-FR-095] Video Comments and Discussion Threads [DEFERRED]

Timestamped comments/discussion on the Video Detail Page. Deferred to Phase 2 per SD-6.

Priority: Future Phase (P4)

Source: SD-6

5. Non-Functional Requirements

5.1 Performance

[VTM-NFR-010] Playback Start Latency

Video playback begins within 3 seconds on 10 Mbps broadband. Adaptive bitrate streaming adjusts dynamically.

Priority: Must-have (P1)

Source: Derived

[VTM-NFR-011] Upload Throughput

Sustain concurrent uploads from 50 instructors to institutional media server without degradation below 80% bandwidth. External platform uploads subject to platform rate limits.

Priority: Should-have (P2)

Source: Derived

[VTM-NFR-012] Search and API Response Time

UI search: within 2 seconds for catalogs up to 100,000 videos. API catalog queries: within 500ms (p95) for standard paginated requests. Video detail: within 200ms (p95). Search index updated within 30 seconds of metadata change.

Priority: Must-have (P1)

Source: Derived; elevated from P2 per SD-5

5.2 Scalability

[VTM-NFR-020] Scaling Targets

Transcoding, streaming, metadata, and API services independently scalable. API gateway sized for 20 concurrent OAuth clients (SD-13) with headroom for growth. No cross-instance federation required (SD-15). CDN-backed delivery for institutional media server.

Priority: Must-have (P1)

Source: Derived; refined per SD-13, SD-15

5.3 Accessibility

[VTM-NFR-030] WCAG 2.1 AA Compliance

All UI meets WCAG 2.1 Level AA. Player fully keyboard-navigable, screen-reader supported (ARIA labels), closed captions, high-contrast mode.

Priority: Must-have (P1)

Source: Derived

[VTM-NFR-031] Audio Description Track

Optional audio description track for visually impaired users.

Priority: Could-have (P3)

Source: Derived

5.4 Security

[VTM-NFR-040] Authentication and Authorization

All access requires Sakai authentication (UI) or OAuth 2.0 token (API). Video stream URLs signed and time-limited (default TTL 4 hours). Signed-URL protection is the sole streaming access control; no DRM (SD-3).

Priority: Must-have (P1)

Source: Derived; refined per SD-3

[VTM-NFR-041] Content Protection

Videos encrypted at rest (AES-256) on institutional media server. Streams delivered over HTTPS/TLS 1.2+. DRM explicitly out of scope (SD-3).

Priority: Should-have (P2)

Source: Derived; simplified per SD-3

[VTM-NFR-042] API Security

OAuth 2.0 token validation on every request. TLS 1.2+ for all API traffic. HMAC-signed webhook payloads. Rate limiting per client. Optional IP allowlisting. API keys and OAuth secrets stored encrypted. All auth failures logged. CORS whitelisting. Input validation. OWASP API Security Top 10 mitigations.

Priority: Must-have (P1)

Source: Derived — API hardening

[VTM-NFR-043] External Platform Credential Security

OAuth 2.0 tokens for YouTube and Vimeo encrypted at rest (AES-256), stored in secrets vault or encrypted DB column, never exposed in logs/API/UI. Automatic refresh before expiry.

Priority: Must-have (P1)

Source: Derived — credential management

5.5 Reliability and Availability

[VTM-NFR-050] Uptime SLA

VTM streaming and browsing: 99.5% uptime. External LMS API: 99.9% uptime. Transcoding and captioning failures retried up to 3 times before instructor notification.

Priority: Must-have (P1)

Source: Derived

5.6 Compatibility

[VTM-NFR-060] Browser and Device Support

Latest two major versions of Chrome, Firefox, Safari, Edge. Responsive and fully functional on tablets and mobile (iOS Safari, Android Chrome).

Priority: Must-have (P1)

Source: Derived

5.7 Internationalization

[VTM-NFR-070] Multi-Language UI

All interface strings externalised in Sakai i18n resource bundles. RTL language support. API error messages include machine-readable code alongside human-readable message.

Priority: Should-have (P2)

Source: Derived

5.8 Data Privacy and Compliance

[VTM-NFR-080] GDPR / FERPA Compliance

Student viewing data treated as PII. Support data export, anonymization, deletion requests. YouTube embeds use youtube-nocookie.com (GDPR). API does not expose student PII unless client has analytics scope. Analytics data anonymizable for aggregated reporting.

Priority: Must-have (P1)

Source: Derived

6. Data Model (Conceptual)

Key entities. Full ER diagrams and schemas elaborated during Design phase.

Entity	Key Attributes
Video	video_id, title, description, duration, format, storage_location (institutional/youtube/vimeo), storage_path, external_platform_video_id, external_embed_url, external_url, thumbnail_url, upload_date, uploader_id, site_id, visibility_scope, status, restriction_flag, publish_date, expiry_date
Video_Course_Allocation	video_id, site_id, allocated_date, allocated_by (many-to-many)
Category	category_id, name, parent_category_id, scope (Global/Course), site_id, sort_order
Video_Category	video_id, category_id (many-to-many)
Playlist	playlist_id, name, site_id, creator_id, description, created_date
Playlist_Item	playlist_id, video_id, sort_order
Caption_Track	track_id, video_id, language_code, file_path, is_auto_generated, external_sync_status
Chapter_Marker	marker_id, video_id, timestamp_seconds, title

Annotation	annotation_id, video_id, timestamp_seconds, text, author_id
View_Event	event_id, video_id, user_id, session_start, session_end, pct_viewed, site_id, source (vtm/api/lesson)
Audit_Log	log_id, video_id, action, actor_id, timestamp, before_value, after_value, source (ui/api)
Quota_Config	config_id, scope (Institution/Dept/Course), scope_id, max_bytes, used_bytes
External_Platform_Config	config_id, platform_type (youtube/vimeo), display_name, client_id, client_secret_encrypted, channel_id, default_privacy, oauth_token_encrypted, token_expiry
API_Client	client_id, client_secret_hash, name, description, contact_email, scope (read/manage/analytics), rate_limit, is_active, created_date, max_20_clients (SD-13)
API-Token	token_id, client_id, access_token_hash, issued_at, expires_at, is_revoked
Webhook_Config	webhook_id, client_id, url, events_subscribed, signing_secret, is_active
Webhook_Delivery_Log	log_id, webhook_id, event_type, payload_hash, http_status, delivered_at, retry_count

7. User Interface and Experience Requirements

7.1 Global Video Page

Grid or list view of published Global-scope videos with thumbnail, title, author, duration, view count, category badges, and storage-location icon (admin/instructor view). Search bar and filter sidebar. Infinite scroll or pagination.

7.2 Course Video Page

Mirrors Global page scoped to current course. Playlists and course-specific category navigation. Instructors see Upload button (with storage target selector) and management toggle.

7.3 Video Detail Page

Player (native or YouTube/Vimeo-embedded), description, metadata, chapters, transcript, related videos, and (for instructors) analytics, management actions, external platform sync status.

7.4 Upload and Edit Dialogs

Modal dialogs or slide-out panels. Required fields clearly marked. Storage Target dropdown: Institutional Media Server and configured YouTube/Vimeo channels. Validation errors inline. Students see no upload controls (SD-10).

7.5 Lessons Integration UI

Lessons content picker includes a 'Video Library' tab. Embedded player matches standalone VTM player. Context menu: 'Open in Video Library' and 'Promote to Video Library'.

7.6 My Videos Dashboard (Student)

Watch Later queue, Favourites, Recently Watched, In Progress with resume. No upload functionality (SD-10).

7.7 Private Channel Administration UI

Admin settings page for managing YouTube and Vimeo connections. Lists configured platforms with status, last sync, video count, actions (edit, test, disconnect). OAuth flow initiated from this page.

7.8 API Client Management UI

Admin page to register, edit, revoke API clients (max 20 per SD-13). Displays: client name, ID, scope (read / read+manage / read+manage+analytics), rate limit, creation date, last used, request count. Client_secret shown once on generation. Webhook config as sub-section per client.

8. Migration and Backward Compatibility (Optional — SD-8)

[VTM-FR-200] Migration of Existing Video Content

Migration utility scans Sakai Resources and Lessons for video files and offers import into VTM with metadata and visibility. Also supports importing from existing YouTube/Vimeo channels.

Priority: Could-have (P3)

Source: Derived — optional per SD-8

[VTM-FR-201] Backward-Compatible Lesson Embeds

Existing video embeds in Lessons (Resources or iframe) shall continue to function after VTM deployment. Zero regression.

Priority: Must-have (P1)

Source: Derived — zero regression

9. Requirements Traceability to Original Inputs

Req #	Description	VTM Requirement IDs
1	Upload videos: public or course-only	VTM-FR-010, VTM-FR-020
2	Videos independent or arising from Lesson usage	VTM-FR-022, VTM-FR-031
3	Lesson upload → publish/keep private; toggle later	VTM-FR-020, 021, 022
4	VTM video embeddable in Lessons if not restricted	VTM-FR-030
5	Videos grouped by courses / categories / subcategories	VTM-FR-050, 051, 052
6	Withdrawal from video page	VTM-FR-023
7	External links for video embedding	VTM-FR-012
8 (v2.0)	Private channel storage on YouTube/Vimeo	VTM-FR-110–116
9 (v2.0)	External LMS API for video catalog access and management	VTM-FR-120–131

10. Prioritization Summary (MoSCoW)

Priority	Count	Requirement IDs
Must-have (P1)	27	VTM-FR-010, 012, 020, 021, 022, 023, 030, 031, 040, 050, 051, 060, 070, 071, 075, 080, 100, 102, 110, 111, 113, 116, 120, 121, 122, 123, 124, 125, 126, 131, 201; VTM-NFR-010, 012, 020, 030, 040, 042, 043, 050, 060, 080
Should-have (P2)	22	VTM-FR-013, 024, 032, 033, 052, 053, 061, 062, 072, 073, 081, 082, 090, 091, 101, 112, 114, 115, 127, 128, 129, 130; VTM-NFR-011, 041, 070
Could-have (P3)	8	VTM-FR-025, 041, 074, 083, 103, 104, 200; VTM-NFR-031
Future Phase (P4)	2	VTM-FR-011 (Bulk Upload), VTM-FR-095 (Video Comments)

11. Open Questions and Assumptions

11.1 Resolved Questions (All)

All open questions from v1.0 through v3.0 have been resolved. The table below is the complete resolution log.

ID	Question	Resolution	Version
OQ-1	Student-generated video uploads?	No. Students cannot upload videos. Upload is Instructor/TA/Admin only.	v3.0
OQ-2	Preferred storage backend?	Institutional media server (primary) + YouTube/Vimeo private channels (optional).	v2.0
OQ-3	Auto-captioning service?	External third-party API (Google, AWS Transcribe, or Whisper).	v2.0
OQ-4	DRM required?	No. Signed-URL protection is sufficient.	v2.0
OQ-5	Live streaming or on-demand?	On-demand only. Live streaming is out of scope.	v2.0
OQ-6	Expected catalog size?	Large. Architecture supports 100,000+ videos with high growth.	v2.0
OQ-7	Video comments/discussion?	Deferred to Phase 2.	v2.0
OQ-8	Which external platforms?	YouTube and Vimeo only. Dailymotion, Wistia, others excluded.	v3.0
OQ-9	API write scope?	Read + manage existing videos (update metadata, change visibility, withdraw/republish). No create, no delete.	v3.0
OQ-10	Max simultaneous API clients?	20	v3.0
OQ-11	GraphQL support?	No. REST is sufficient.	v3.0
OQ-12	Multi-instance federation?	No. Single Sakai instance only.	v3.0

11.2 Remaining Open Questions

None. All stakeholder questions have been resolved as of v3.0.

11.3 Assumptions

- A-1: The Sakai instance is version 25.x or later with the Lessons tool enabled (SD-9).

- A-2: A CDN or reverse proxy is available for video delivery to reduce origin load.
- A-3: Server-side transcoding infrastructure (e.g., FFmpeg cluster) is provisioned alongside the institutional media server.
- A-4: The institution has existing authentication via CAS/SAML integrated with Sakai.
- A-5: The institution has or will obtain API credentials for YouTube Data API v3 and Vimeo API.
- A-6: External LMS platforms connecting via the API can integrate with OAuth 2.0 and consume REST/JSON endpoints.
- A-7: The institutional media server provides sufficient bandwidth and storage for the expected large catalog; YouTube/Vimeo channels serve as optional secondary storage, not primary.
- A-8 (v3.0): The 20 external LMS API clients are sufficient for the institution's current integration needs; the limit is configurable and can be raised in a future release.

Appendix A: Video Lifecycle State Diagram

State transitions for a video object. For Private-Channel-Hosted videos, transitions trigger privacy status changes on YouTube/Vimeo (see VTM-FR-113).

- Draft: Initial state after upload. Transcoding or awaiting metadata completion. Not visible to students. Platform: private.
- Published: Live and visible per Visibility Scope. May transition to Withdrawn or Archived. Platform: unlisted.
- Withdrawn: Hidden from browsing/search/API catalog. May still play in existing Lesson embeds. Can be republished. Platform: private. Triggerable via External LMS API (manage scope).
- Archived: Moved to cold storage (institutional) or marked inactive (YouTube/Vimeo). Restorable to Draft.
- Pending Approval: Intermediate state for Global-scope videos when moderation workflow enabled. Transitions to Published (approved) or Draft (rejected).

Appendix B: External LMS API — Endpoint Summary

Δ v3.0: Updated to reflect read + manage scope (SD-12). Create/delete endpoints removed.

Method	Endpoint	Description	Auth / Scope
POST	/api/v1/oauth/token	Obtain access token (Client Credentials grant)	client_id / secret
GET	/api/v1/videos	List/search videos (paginated, filterable, sortable)	Bearer / read
GET	/api/v1/videos/{id}	Single video full detail incl. course allocations, categories	Bearer / read
PUT	/api/v1/videos/{id}	Update video metadata (title, description, tags, categories, courses, custom_metadata)	Bearer / manage

PATCH	/api/v1/videos/{id}/visibility	Change visibility scope	Bearer / manage
PATCH	/api/v1/videos/{id}/status	Publish, withdraw, or republish a video	Bearer / manage
PUT	/api/v1/videos/{id}/categories	Replace category assignments	Bearer / manage
PUT	/api/v1/videos/{id}/courses	Replace course allocations	Bearer / manage
PATCH	/api/v1/videos/{id}/schedule	Update publish/expiry dates	Bearer / manage
POST	/api/v1/videos/{id}/embed	Generate signed embed URL	Bearer / read
GET	/api/v1/courses/{id}/videos	Videos for a specific course	Bearer / read
GET	/api/v1/courses/{id}/analytics	Course-level video analytics	Bearer / analytics
GET	/api/v1/categories	Category taxonomy hierarchy	Bearer / read
GET	/api/v1/playlists	List playlists	Bearer / read
GET	/api/v1/playlists/{id}	Playlist detail with ordered video list	Bearer / read
GET	/api/v1/health	API health / status check	Public (no auth)
GET	/api/v1/openapi.json	Machine-readable OpenAPI 3.1 specification	Public (no auth)

Appendix C: Glossary

- CDN: Content Delivery Network — geographically distributed servers for low-latency video delivery.
- HLS: HTTP Live Streaming — Apple’s adaptive streaming protocol.
- DASH: Dynamic Adaptive Streaming over HTTP — international standard for adaptive streaming.
- oEmbed: Open format for embedding content from a URL, used by YouTube, Vimeo, and others.
- xAPI: Experience API (Tin Can) — specification for learning analytics data collection.
- OAuth 2.0: Open standard for access delegation; used for YouTube/Vimeo API access and External LMS API client authentication.
- OpenAPI 3.1: Specification for describing RESTful APIs (formerly Swagger).
- HMAC-SHA256: Hash-based Message Authentication Code used for signing webhook payloads and embed URLs.
- Webhook: HTTP callback delivering real-time event notifications to a registered URL.
- VOD: Video on Demand — content available for playback at the viewer’s convenience (as opposed to live streaming).

— End of Document —